

First Pattern Failure in Catalan Permutations

A proof of OEIS A273821, exact formulas,
all-orders expansions, and a critical phase transition

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Abstract

For a uniformly chosen 123-avoiding permutation of $[n]$, let K_n be the largest number of its largest entries whose induced permutation still avoids 132. The generating function for the joint counts $T(n, k)$ is marked conjectural in OEIS A273821. We prove it by inserting successive minima, then telescope the resulting Catalan convolution to obtain a difference of two binomial coefficients. This resolves both the posted generating-function conjecture and the indicated growth-rate question for every nonzero fixed column.

The exact formula yields substantially more: every column $k \geq 2$ is hypergeometric in n ; its full inverse-power expansion is computable to any order; and K_n has an explicit discrete limit law with exponentially weighted total-variation error $O(n^{-1})$. The fixed-column approximation changes at $k \asymp \sqrt{n}$, where we prove an explicit crossover. For k proportional to n , two distinct exponential sectors are separated exactly rather than hidden under an algebraic error term.

Weighting permutations by y^{K_n} produces a transition at $y = 2$. Below the transition K_n stays bounded; above it $n - K_n$ stays bounded; at the transition K_n/n converges to $\text{Beta}(1, \frac{1}{2})$. We give an exact critical partition function and mean, a critical window, Lambert- W inverse expansions, independent computation, and a formalization and research agenda.

Status. The report supplies conventional proofs and executable checks, not a Lean formalization. The identified OEIS conjecture is proved here; priority for every extension is not certified by an exhaustive literature search. No OEIS edit or repository change has been submitted.

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1 The question, the answer, and the scope of the claim

1.1 The statistic and the live OEIS question

A permutation avoids a pattern if no subsequence, with positions increasing, has the same relative order as that pattern. Thus 123 avoidance forbids three increasing entries, and 132 avoidance forbids indices $i < j < h$ with $\pi_i < \pi_h < \pi_j$. Write $\text{Av}_n(123)$ for the first class.

For $\pi \in \text{Av}_n(123)$, restrict π to its k largest values without changing their order. Let $K(\pi)$ be the largest k for which this restriction avoids 132. Set

$$T(n, k) = \#\{\pi \in \text{Av}_n(123) : K(\pi) = k\}, \quad Z_n(y) = \sum_{k=1}^n T(n, k)y^k. \quad (1.1)$$

The triangle is A273821, introduced by David Callan in 2016 [1]. As retrieved on October 1, 2026, its bivariate generating function remains labeled conjectural; its comments also suggest growth constant 4 for each column other than the first. These are the two specific posted questions answered below.

We use $C_n = \frac{1}{n+1} \binom{2n}{n}$ and

$$C(x) = \sum_{n \geq 0} C_n x^n = \frac{2}{1 + \sqrt{1 - 4x}}, \quad C(x) = 1 + xC(x)^2. \quad (1.2)$$

Our insertion argument will itself establish the Catalan row sums, so the proof does not assume the conjectural triangle recurrence or generating function. All entries outside $1 \leq k \leq n$ are zero. For a binomial coefficient with nonnegative upper argument, a negative lower argument is interpreted as zero.

1.2 Principal conclusions

The central enumeration result is

$$T(n, k) = 2^{k-1} \binom{2n-2k}{n-k-1} - \binom{2n-k-1}{n-k-1} \quad (n > k \geq 1), \quad (1.3)$$

with the separate diagonal value $T(n, n) = 2^{n-1}$. In particular, the first column is zero after its initial entry.

The resulting bivariate generating function is

$$F(x, y) := \sum_{n \geq 1} Z_n(y)x^n = \frac{xy}{1-2xy} + \frac{x^3 y^2 C(x)^3}{(1-2xy)(1-xyC(x))}. \quad (1.4)$$

This is algebraically the conjectured OEIS expression. Its present form records the first failure of 132 avoidance explicitly.

For every fixed $k \geq 2$,

$$T(n, k) \sim \frac{(k-1)(k+4)}{2^{k+3}\sqrt{\pi}} \frac{4^n}{n^{3/2}}. \quad (1.5)$$

We prove a full expansion with explicit first coefficients and an all-orders recursion, not only this leading equivalent. Moreover, for a uniform permutation in $\text{Av}_n(123)$,

$$\mathbb{P}(K_n = k) \longrightarrow p_k := \frac{(k-1)(k+4)}{2^{k+3}} \quad (k \geq 2), \quad \sum_{k \geq 2} p_k y^k = \frac{y^2(3-y)}{2(2-y)^3}. \quad (1.6)$$

There is a second, macroscopic phenomenon. Under the probability measure proportional to $y^{K(\pi)}$, the scaling of K_n changes at $y = 2$:

$$\begin{array}{l|l} 0 < y < 2 & K_n \text{ converges to an explicit discrete law,} \\ y = 2 & K_n/n \Rightarrow \text{Beta}(1, \frac{1}{2}), \\ y > 2 & n - K_n \text{ converges to an explicit discrete law.} \end{array} \quad (1.7)$$

The critical normalization has the exact evaluation

$$Z_n(2) = (n+2) \binom{2n}{n} - 4^n. \quad (1.8)$$

These conclusions are consequences of the same exact first-failure formula, not independent guesses from numerical plots.

1.3 What is established, and what is not claimed

The contribution is a self-contained resolution of an identifiable posted conjecture, together with a detailed analysis of its refined counting triangle. Pattern-avoiding permutations, Catalan generating trees, Stirling expansions, and singularity analysis are established methods; they are not claimed as new here. The two low columns already have known companion entries, [A000245](#) and [A071718](#), including known leading asymptotics [2, 3].

A targeted search of the sequence identifier and defining statistic did not locate an independent published proof of the A273821 conjecture. That is a limited search result, not a proof of priority. Accordingly, “proved here” is used literally, while an assertion that every extension is unprecedented is deliberately avoided.

The ProveIt repository has an independent transseries library for asymptotic scales and differential blocks, together with an inversion volume, a combinatorial companion, and verification programs [9]. Here we distinguish formal expansions, analytic interpretation, and executable checks, especially when an exponentially smaller sector is present. No theorem from the repository is needed as an unverified premise, and none of the results below is represented as already formalized in Lean.

1.4 Provenance and place in the collection

[write] Added 1 October 2026, batch 74. This report is a single manuscript, printed in full: manuscript 04 of batch 74 of ProveIt’s incoming-reports intake, delivered as the archive `A273821_First_Pattern_Failure.zip` in commit `c664fc2f0` and placed in the research-report collection (`SetTheory/Cardinals/docs/reports/enumerative-combinatorics/`) in commit `b669cff87`. Its pin is commit `291fbe6bb`, at which its author read the root README and, from the default branch, `Analysis/Transseries/README.md` (blob `3d9ce016`). The writing step changed only the labels (every label now carries the prefix `fpf:`), added these **[write]** notes and six section labels, and added two repository references [7, 8]; no statement or proof was altered.

No other report in the collection treats A273821 or its statistic. The nearest relatives are of the same genre only: `enumerative-combinatorics/mesh-avoidance-catalan-inflation` proves a Catalan-type generating function marked conjectural in OEIS A289587, and `enumerative-combinatorics/adjacency-bounded-132-avoiders` studies 132-avoiders under a different constraint. Neither shares a theorem with this report.

The transseries material mentioned above remains methodological context only. The inverse expansion of Section 11 is an instance of the balanced inverse theorem of the combinatorial-inverses volume [7], and the rounding discussion is an instance of the staircase theorem of the transseries-and-inversion volume [8]; neither is claimed as new. Placement in the collection, beside the repository’s Lean and Rocq developments, confers no formal status: nothing in this report is formalized.

2 A minimum-insertion model for Catalan permutations

2.1 The active suffix

Every nonempty permutation has a unique parent obtained by deleting its minimum and standardizing the remaining values. Reversing this operation builds a permutation by inserting successively smaller values.

Definition 2.1. The suffix state d of a nonempty permutation is the length of its longest terminal decreasing subsequence of *consecutive positions*. Thus the last d entries decrease, and either the permutation is decreasing or the entry immediately before them forms an ascent with the first of those d entries. An insertion site is indexed by the number j of old entries to its right.

Lemma 2.2 (Active sites). *For a permutation in $\text{Av}_n(123)$ with suffix state d , insertion of a new minimum preserves 123 avoidance exactly at the sites $0 \leq j \leq d$. The new state is $d + 1$ when $j = 0$, and is j when $1 \leq j \leq d$. Consequently the state production rule is*

$$d \longrightarrow 1, 2, \dots, d, d + 1, \quad (2.1)$$

with each child state occurring once.

Proof. Any newly formed increasing triple must start at the new minimum. Such a triple exists precisely when the entries to its right contain an increasing pair. Avoiding such a pair means that this right-hand suffix is decreasing, which is equivalent to $j \leq d$. If the minimum is appended, it lengthens the decreasing suffix by one. Otherwise it lies just before a decreasing suffix of length j and is smaller than its first entry, so that suffix cannot extend to the left. Its new length is exactly j . $\square$

2.2 Descendant generating functions

Let $D_d(x)$ count descendants of a state- d permutation, with x marking the number of additional minima inserted. The particular values or earlier shape of the permutation do not matter, because the production rule depends only on d .

Lemma 2.3. *For every $d \geq 1$,*

$$D_d(x) = C(x)^{d+1}. \quad (2.2)$$

Proof. The production rule gives

$$D_d(x) = 1 + x \sum_{j=1}^{d+1} D_j(x).$$

This infinite system has a unique formal-power-series solution with constant term 1: after the coefficients through degree $m - 1$ have been determined for every state, the equation determines the coefficient of degree m for every state using only a finite sum. To verify the claimed solution, use $xC^2 = C - 1$:

$$x \sum_{j=1}^{d+1} C^{j+1} = (C - 1)(1 + C + \dots + C^d) = C^{d+1} - 1.$$

Uniqueness finishes the proof. $\square$

The single-entry permutation has state 1. Therefore the number of size- n permutations generated is

$$[x^{n-1}]D_1(x) = [x^{n-1}]C(x)^2 = C_n,$$

where the final equality follows from $xC^2 = C - 1$. Thus this construction also proves the Catalan row sums.

2.3 The two safe sites for avoiding both patterns

Suppose the current permutation avoids both 123 and 132. At an active site the entries to the right of a newly inserted minimum decrease. The new minimum starts a 132 occurrence exactly when there are at least two such entries. Hence the insertions that remain safe for both patterns are precisely $j = 0$ and $j = 1$.

The first appends the new minimum and increases d by one; the second places it before the last entry and resets d to 1. These two choices encode all size- k safe prefixes by binary words of length $k - 1$.

Lemma 2.4. *Let $a_{k,d}$ be the number of permutations of size k avoiding both patterns and having suffix state d . Then*

$$a_{k,k} = 1, \quad a_{k,d} = 2^{k-d-1} \quad (1 \leq d < k). \quad (2.3)$$

In particular, there are 2^{k-1} such permutations, and for $2 \leq j \leq k$,

$$\sum_{d=j}^k a_{k,d} = 2^{k-j}. \quad (2.4)$$

Proof. The state is k only if every insertion appends. Otherwise locate the last reset: to finish at state $d < k$ it must be followed by exactly $d - 1$ appends. The preceding $k - d - 1$ decisions are free. Summation gives both remaining assertions. $\square$

3 First failure and the conjectured generating function

3.1 The first-failure decomposition

Once a 132 occurrence has appeared among the largest entries, inserting smaller entries cannot remove it. Thus $K = k < n$ is equivalent to remaining safe through size k and failing at the next insertion. At that failure insertion the active site has $j \geq 2$ entries to its right.

Theorem 3.1 (Positive Catalan convolution). *For $n > k \geq 2$,*

$$T(n, k) = \sum_{j=2}^k 2^{k-j} [x^{n-k-1}] C(x)^{j+1}. \quad (3.1)$$

For $n = k$, $T(n, n) = 2^{n-1}$; for $n > 1$, $T(n, 1) = 0$.

Proof. For a fixed failing site j , a safe size- k prefix permits that insertion exactly when its state d satisfies $d \geq j$. There are 2^{k-j} such prefixes by Lemma 2.4. The failing insertion gives state j by Lemma 2.2; the remaining $n - k - 1$ insertions are arbitrary 123-preserving descendants and are counted by C^{j+1} . The parent operation is unique, so these cases are disjoint and exhaustive. The boundary cases follow from the safe-prefix count and the fact that a 132 occurrence needs three entries. $\square$

For example, the row at size 5 is

$$(T(5, 1), \dots, T(5, 5)) = (0, 9, 10, 7, 16).$$

The off-diagonal formula gives $T(5, 3) = [x](2C^3 + C^4) = 6 + 4 = 10$. The row sums to $42 = C_5$. This example is computed from the insertion argument, not used to infer it.

3.2 Summation of the columns

Theorem 3.2 (Proof of the posted generating-function conjecture). *The identity (1.4) holds in $\mathbb{Q}[y][[x]]$. It is equal to*

$$C(x) - 1 + \frac{(1-y)(1-xy)\{1 - (1-xy)C(x)\}}{(1-2xy)(1-y+xy^2)}, \quad (3.2)$$

the expression marked conjectural in OEIS A273821 [1].

Proof. The size generating function for column k is

$$G_k(x) = 2^{k-1}x^k + x^{k+1} \sum_{j=2}^k 2^{k-j} C(x)^{j+1}, \quad G_1(x) = x. \quad (3.3)$$

The diagonal contribution sums to $xy/(1-2xy)$. In the second contribution, write $k = j + h$ and sum the two geometric series:

$$\begin{aligned} \sum_{k \geq 2} y^k x^{k+1} \sum_{j=2}^k 2^{k-j} C^{j+1} &= xC \sum_{j \geq 2} (xyC)^j \sum_{h \geq 0} (2xy)^h \\ &= \frac{x^3 y^2 C^3}{(1-2xy)(1-xyC)}. \end{aligned}$$

This proves (1.4). To compare with (3.2), replace xC^2 by $C - 1$ and use

$$1 - y + xy^2 = (1 - xyC)(1 - y/C).$$

Clearing denominators then gives identical numerators. This is an identity of rational algebraic functions; any removable factors in the displayed OEIS form are interpreted through the formal series (1.4). $\square$

Setting $y = 1$ gives $F(x, 1) = C(x) - 1$, consistently with the combinatorial row-sum proof. The derivation also shows why two apparently different singular factors occur: $1 - 2xy$ records the binary safe-prefix evolution, whereas $1 - xyC$ records the active suffix at first failure.

4 Two binomials, exact tails, and hypergeometric columns

4.1 A telescoping Catalan convolution

Lagrange inversion applied to $C = 1 + xC^2$ gives, for $m \geq 0$ and $j \geq 1$,

$$b(m, j) := [x^m]C(x)^{j+1} = \frac{j+1}{2m+j+1} \binom{2m+j+1}{m} = \binom{2m+j}{m} - \binom{2m+j}{m-1}. \quad (4.1)$$

For completeness, if $m > 0$, set $u = C - 1 = x(1+u)^2$. The Lagrange formula for $[x^m](1+u)^{j+1}$ is $\frac{j+1}{m} [u^{m-1}](1+u)^{2m+j}$, which simplifies to the first expression. At $m = 0$ both sides are 1.

Lemma 4.1 (Telescopier). *For $t_j = 2^{-j} \binom{2m+j}{m}$,*

$$2^{-j} b(m, j) = 2(t_j - t_{j+1}). \quad (4.2)$$

Proof. Pascal's identity gives

$$2 \binom{2m+j}{m} - \binom{2m+j+1}{m} = \binom{2m+j}{m} - \binom{2m+j}{m-1}.$$

Multiplication by 2^{-j} is exactly the claim. $\square$

Theorem 4.2 (Exact entries and cumulative counts). *The binomial formula (1.3) holds. In addition, for $1 \leq k \leq n$,*

$$S(n, k) := \sum_{h=k}^n T(n, h) = 2^{k-1} \left(\binom{2n-2k+1}{n-k} - \binom{2n-k}{n-k-1} \right). \quad (4.3)$$

Proof. In (3.1) put $m = n - k - 1$. Lemma 4.1 telescopes the sum to $2^{k+1}(t_2 - t_{k+1})$, giving (1.3). For $k = 1 < n$, its two terms coincide, as required.

For the cumulative count, being safe through size k is the only restriction. Put $\ell = n - k$. Lemmas 2.3 and 2.4 give

$$S(n, k) = [x^\ell] \left(C^{k+1} + \sum_{d=1}^{k-1} 2^{k-d-1} C^{d+1} \right).$$

With $m = \ell$, the telescoping sum over d is $2^k(t_1 - t_k)$, and the remaining term is $b(\ell, k)$. The result is $2^{k-1} \binom{2\ell+1}{\ell} - \binom{2\ell+k}{\ell-1}$, which is (4.3). It includes $k = n$ under the stated binomial convention. $\square$

In particular $S(n, 1) = C_n$. The positive formula (3.1) proves positivity of the difference of binomials for $2 \leq k < n$, without needing a cancellation estimate.

4.2 A polynomial factor and a first-order recurrence

Definition 4.3. For $k \geq 2$, define

$$H_k(n) = 2^{k-1} (n)_{\underline{k-1}} - (2n-k-1)_{\underline{k-1}}, \quad (a)_{\underline{r}} = \prod_{j=0}^{r-1} (a-j). \quad (4.4)$$

Theorem 4.4. *The polynomial H_k has degree $k-2$, with leading coefficient $2^{k-3}(k-1)(k+4)$. For $n > k$,*

$$T(n, k) = \frac{(2n-2k)!}{(n-k-1)! n!} H_k(n). \quad (4.5)$$

Consequently each fixed off-diagonal column is hypergeometric in n , with recurrence

$$\begin{aligned} (n-k)(n+1)H_k(n)T(n+1, k) \\ = 2(n-k+1)(2n-2k+1)H_k(n+1)T(n, k), \quad n > k. \end{aligned} \quad (4.6)$$

Proof. Factor the factorial expression in (1.3) to obtain (4.5). Both terms of H_k have leading term $2^{k-1}n^{k-1}$, so those terms cancel. Expanding the next coefficient by summing the roots of the two products gives $2^{k-3}(k-1)(k+4) > 0$. Thus the degree is exactly $k-2$. Taking the ratio of the factorial prefactors at $n+1$ and n gives (4.6). Positivity follows from Theorem 3.1 and the positive prefactor. $\square$

The first examples are

$$H_2(n) = 3, \quad H_3(n) = 14n - 20, \quad H_4(n) = 48n^2 - 198n + 210. \quad (4.7)$$

This first-order structure is stronger than merely knowing that each column is D-finite.

4.3 Known companion sequences and their offsets

The identifications noted in the OEIS entry become exact with the following ranges:

$$\begin{aligned} T(n, 2) &= A000245(n-2), & n &\geq 3, \\ T(n, 3) &= A071718(n-3), & n &\geq 4. \end{aligned} \quad (4.8)$$

For the second column, (3.3) gives $G_2 = 2x^2 + x^3C^3$; the companion generating function is xC^3 . For the third, $G_3 = 4x^3 + x^4(2C^3 + C^4)$, and

$$1 + x(2C^3 + C^4) = C^3 + x^2C^4 = (1 + x^2C)C^3,$$

which is the companion generating function. These identities follow using $xC^2 = C - 1$. The initial diagonal exceptions are important: $T(2, 2) = 2$ instead of the companion's initial 0, and $T(3, 3) = 4$ instead of its initial 1. The two companion leading equivalents already appear in their entries and are not novelty claims of this report [2, 3].

5 All-orders fixed-column asymptotics

5.1 A rational ratio to the Catalan numbers

Proposition 5.1. *For each $k \geq 2$ and $n > k$, the ratio $T(n, k)/C_n$ is rational in n :*

$$\begin{aligned} \frac{T(n, k)}{C_n} &= (n+1) \frac{2^{k-1} (n)_{\underline{k+1}} (n)_{\underline{k-1}}}{(2n)_{\underline{2k}}} \\ &\quad - (n+1) \frac{(n)_{\underline{k+1}}}{(2n)_{\underline{k+1}}}. \end{aligned} \quad (5.1)$$

Proof. Divide each binomial term of (1.3) by $(2n)!/(n!^2(n+1))$ and cancel the factorials. $\square$

The two terms on the right individually grow linearly with n for fixed k , but their difference has a finite limit. Resolving that cancellation before expanding is useful both algebraically and numerically.

5.2 An explicit coefficient engine

Put $t = 1/n$ and $S_m(q) = \sum_{r=0}^q r^m$. For $m \geq 1$ define

$$\begin{aligned} a_m(k) &= \frac{2^{-m} S_m(2k-1) - S_m(k) - S_m(k-2)}{m}, \\ b_m(k) &= \frac{(2^{-m} - 1) S_m(k)}{m}, \end{aligned} \quad (5.2)$$

and let $A_k(t) = \sum_{m \geq 1} a_m(k) t^m$, $B_k(t) = \sum_{m \geq 1} b_m(k) t^m$. Taking logarithms of the finite products in (5.1) yields the local identity

$$\frac{T(n, k)}{C_n} = 2^{-k-1} \frac{1+t}{t} (e^{A_k(t)} - e^{B_k(t)}). \quad (5.3)$$

For fixed k , this is the Taylor expansion of a rational function at $t = 0$ after a removable singularity is canceled. The series therefore converges for sufficiently small $|t|$, unlike the separate Stirling series used for C_n .

The leading difference is

$$d_1(k) := a_1(k) - b_1(k) = \frac{(k-1)(k+4)}{4}. \quad (5.4)$$

Exponentials of either logarithmic series can be generated without repeated symbolic exponentiation: if $E(t) = \exp(\sum_{m \geq 1} \alpha_m t^m) = \sum E_j t^j$, then

$$E_0 = 1, \quad E_j = \frac{1}{j} \sum_{m=1}^j m \alpha_m E_{j-m}. \quad (5.5)$$

This supplies every probability correction as a rational function of k .

To include the Catalan prefactor, write the formal Stirling logarithm

$$L_C(t) = -\log(1+t) + \sum_{r \geq 1} \frac{B_{2r}(2^{1-2r} - 2)}{2r(2r-1)} t^{2r-1}, \quad (5.6)$$

where B_{2r} are Bernoulli numbers with $B_2 = 1/6$. Stirling's expansion [5] gives

$$C_n \sim \frac{4^n}{\sqrt{\pi n^{3/2}}} e^{L_C(1/n)} = \frac{4^n}{\sqrt{\pi n^{3/2}}} \left(1 - \frac{9}{8n} + \frac{145}{128n^2} - \frac{1155}{1024n^3} + \cdots \right). \quad (5.7)$$

The symbol $\sim$ here means agreement at every fixed truncation order; it does not assert convergence of the infinite Stirling series.

Theorem 5.2 (Full fixed-column expansion). *For every fixed integer $k \geq 2$ and every $J \geq 0$,*

$$T(n, k) = \frac{p_k}{\sqrt{\pi}} \frac{4^n}{n^{3/2}} \left(\sum_{j=0}^J \frac{c_j(k)}{n^j} + O_{k,J}(n^{-J-1}) \right), \quad p_k = \frac{(k-1)(k+4)}{2^{k+3}}, \quad (5.8)$$

where $c_0 = 1$ and the formal generating series of the coefficients is

$$\sum_{j \geq 0} c_j(k) t^j = e^{L_C(t)} \frac{(1+t)(e^{A_k(t)} - e^{B_k(t)})}{d_1(k)t}. \quad (5.9)$$

In particular,

$$\begin{aligned} c_1(k) &= -\frac{k^3 - k^2 - 17k + 36}{8(k+4)}, \\ c_2(k) &= \frac{4k^5 - 32k^4 - 22k^3 + 578k^2 - 2601k + 1740}{384(k+4)}. \end{aligned} \quad (5.10)$$

Proof. Equations (5.3) and (5.4) give the convergent fixed- k ratio expansion with leading coefficient $2^{-k-1}d_1 = p_k$. Multiply it by the Stirling expansion (5.7). A finite Taylor remainder for the rational ratio and the finite-order Stirling remainder give the stated error for each fixed J . Applying (5.5) gives (5.10); the derivation is also checked symbolically in the supplied program. $\square$

Both natural meanings of the OEIS growth-rate comment now follow:

$$\lim_{n \rightarrow \infty} T(n, k)^{1/n} = 4, \quad \lim_{n \rightarrow \infty} \frac{T(n+1, k)}{T(n, k)} = 4 \quad (k \geq 2). \quad (5.11)$$

For reference, the corresponding normalized-probability expansion starts

$$\frac{T(n, k)}{C_n} = p_k \left(1 + \frac{r_1(k)}{n} + \frac{r_2(k)}{n^2} + O_k(n^{-3}) \right), \quad (5.12)$$

where

$$\begin{aligned} r_1(k) &= -\frac{k(k^2 - k - 26)}{8(k + 4)}, \\ r_2(k) &= \frac{k(k^4 - 8k^3 - 19k^2 + 158k - 408)}{96(k + 4)}. \end{aligned} \quad (5.13)$$

The dependence on k already warns that these fixed-column expansions cannot be used uniformly all the way across the triangle.

6 The discrete limiting law and an exponentially weighted error

Theorem 6.1. *Let K_n be defined under the uniform measure on $\text{Av}_n(123)$. It converges in distribution to K with probabilities p_k in (1.6). Its probability generating function, mean, variance, and tail are*

$$\begin{aligned} P(y) &= \mathbb{E}y^K = \frac{y^2(3-y)}{2(2-y)^3}, & |y| < 2, \\ \mathbb{E}K &= \frac{9}{2}, & \text{Var}(K) = \frac{21}{4}, \\ \mathbb{P}(K \geq m) &= 2^{-m-2}(m^2 + 5m + 2), & m \geq 1. \end{aligned} \quad (6.1)$$

For every fixed $1 < r < 2$,

$$\sum_{k \geq 1} r^k |\mathbb{P}(K_n = k) - p_k| = O_r(n^{-1}), \quad (6.2)$$

where $p_1 = 0$ and the finite- n probabilities are zero for $k > n$. Consequently all moments converge, with error $O(n^{-1})$ for each fixed moment.

Proof. Summing the quadratic times geometric expression for p_k gives $P(y)$, with $P(1) = 1$. Derivatives at $y = 1$ give the two moments. Summing the same geometric tail gives the last formula in (6.1).

For the error assertion, set $z = \sqrt{1 - 4x}$ in the exact generating function. It becomes

$$F(x, y) = -\frac{y(z-1)(yz^2 - yz + z + 1)}{(yz - y + 2)(yz^2 - y + 2)}. \quad (6.3)$$

Fix $R < 2$. Uniformly for $|y| \leq R$, neither denominator in (1.4) vanishes for $|x| \leq 1/4$: indeed $|2xy| \leq R/2 < 1$ and $|xyC(x)| \leq R/2 < 1$. By compactness the function admits a common slit, dented neighborhood of its dominant square-root singularity. Its expansion at $z = 0$ has coefficient $-2P(y)$ on z . The next nonanalytic term is of order z^3 ; retaining further terms gives uniform coefficient remainders. Standard square-root coefficient transfer [4, Chapter VI] therefore gives

$$\sup_{|y| \leq R} \left| \frac{Z_n(y)}{C_n} - P(y) \right| = O_R(n^{-1}). \quad (6.4)$$

One may also obtain the transfer constants from $[x^n](1-4x)^\alpha = 4^n \Gamma(n-\alpha)/(\Gamma(-\alpha)\Gamma(n+1))$ for nonintegral α ; the uniform analytic remainder is handled on the same dented contour.

Now choose $r < R < 2$. Cauchy's coefficient bound applied in the variable y to (6.4) gives $|\mathbb{P}(K_n = k) - p_k| \leq M_R n^{-1} R^{-k}$. Sum the convergent geometric majorant $(r/R)^k$ to prove (6.2). Polynomial moment weights are bounded by a constant times r^k , which proves moment convergence and its rate. $\square$

The exponential weight is useful: an unweighted pointwise limit alone would not justify interchanging arbitrary moment sums and limits. Here it is justified with a quantitative bound.

7 The square-root crossover and uniform moving-column regimes

7.1 An exact positive factorization

For $n > k \geq 2$, put $\ell = n - k$ and write

$$\begin{aligned} A_{n,k} &= 2^{k-1} \binom{2\ell}{\ell-1} = 2^{k-1} \ell C_\ell, \\ B_{n,k} &= \binom{2\ell+k-1}{\ell-1}, \quad T(n,k) = A_{n,k} - B_{n,k} = A_{n,k}(1 - R_{n,k}). \end{aligned} \quad (7.1)$$

The ratio has the exact product

$$R_{n,k} = \frac{B_{n,k}}{A_{n,k}} = \prod_{j=1}^{k-1} \frac{1 + j/(2\ell)}{1 + (j+1)/\ell} = \prod_{j=1}^{k-1} \left(1 - \frac{j+2}{2(\ell+j+1)} \right). \quad (7.2)$$

All factors lie strictly between zero and one.

Lemma 7.1. *For every $n > k \geq 2$,*

$$0 < R_{n,k} \leq \exp \left(-\frac{(k-1)(k+4)}{4n} \right). \quad (7.3)$$

If $k/\ell \rightarrow 0$, then

$$\log R_{n,k} = -\frac{(k-1)(k+4)}{4\ell} + O\left(\frac{k^3}{\ell^2}\right). \quad (7.4)$$

The last error is uniform when k/ℓ is sufficiently small.

Proof. Use $\log(1-u) \leq -u$ in (7.2), the bound $\ell + j + 1 \leq n$, and $\sum_{j=1}^{k-1} (j+2) = (k-1)(k+4)/2$. For (7.4), expand $\log(1 + j/(2\ell)) - \log(1 + (j+1)/\ell)$. The first-order terms sum to the displayed expression; the sum of the quadratic remainder bounds is $O(k^3/\ell^2)$. $\square$

Theorem 7.2 (Three uniform regimes). *The following assertions hold.*

(i) *If $2 \leq k = o(\sqrt{n})$, then uniformly in this regime,*

$$\frac{T(n,k)}{C_n} = p_k \left(1 + O\left(\frac{k^2}{n}\right) \right). \quad (7.5)$$

(ii) *If $k/\sqrt{n} \rightarrow s \in (0, \infty)$, then*

$$T(n,k) \sim \frac{4^n 2^{-k}}{2\sqrt{\pi n}} \left(1 - e^{-s^2/4} \right). \quad (7.6)$$

Equivalently,

$$\frac{T(n, k)/C_n}{p_k} \longrightarrow \frac{4(1 - e^{-s^2/4})}{s^2}. \quad (7.7)$$

The equivalents are uniform on compact subintervals of $0 < s < \infty$.

(iii) If $k/\sqrt{n} \rightarrow \infty$ and $\ell = n - k \rightarrow \infty$, then for each fixed J ,

$$T(n, k) = \frac{4^n 2^{-k}}{2\sqrt{\pi\ell}} \left(\sum_{j=0}^J \frac{\gamma_j}{\ell^j} + O_J(\ell^{-J-1}) \right) (1 - R_{n,k}), \quad (7.8)$$

where $\gamma_0 = 1$, $\gamma_1 = -9/8$, $\gamma_2 = 145/128$, and the exponentially small ratio is bounded by (7.3).

Proof. The factor $A_{n,k}$ has the uniform Catalan expansion

$$A_{n,k} = \frac{4^n 2^{-k}}{2\sqrt{\pi\ell}} \left(1 - \frac{9}{8\ell} + O(\ell^{-2}) \right).$$

In the first regime, (7.4) and the expansion of $1 - e^{-u}$ give

$$1 - R_{n,k} = \frac{(k-1)(k+4)}{4\ell} \left(1 + O(k^2/n) \right).$$

Dividing by C_n and using $\ell/n = 1 - k/n$ proves (7.5). In the second regime $\ell/n \rightarrow 1$ and $R_{n,k} \rightarrow e^{-s^2/4}$ uniformly on the stated compact sets. This proves (7.6), and comparison with $p_k \sim k^2/2^{k+3}$ proves (7.7). In the third regime, the bound on $R_{n,k}$ tends to zero while the Catalan expansion at ℓ can be taken to any fixed order. Retaining $1 - R_{n,k}$ exactly yields the stated error without confusing algebraic and exponential orders. $\square$

8 Two exponential sectors and the near-diagonal regime

8.1 Linear rays

Suppose $\beta = (n - k)/n$ remains in a compact subset of $(0, 1)$. Define

$$\begin{aligned} h_0(\beta) &= (1 + \beta) \log 2, \\ h_1(\beta) &= (1 + \beta) \log(1 + \beta) - \beta \log \beta, \\ J(\beta) &= h_0(\beta) - h_1(\beta) > 0. \end{aligned} \quad (8.1)$$

The exact decomposition (7.1) gives two separately expandable exponential sectors. If $C(t) = e^{L_C(t)}$ denotes the formal Catalan correction series, then

$$A_{n,k} \sim \frac{e^{nh_0(\beta)}}{2\sqrt{\pi\beta n}} C\left(\frac{1}{\beta n}\right), \quad (8.2)$$

$$B_{n,k} \sim \sqrt{\frac{\beta}{2\pi(1+\beta)n}} e^{nh_1(\beta)} e^{L_\beta(1/n)}, \quad (8.3)$$

where

$$L_\beta(t) = \sum_{r \geq 1} \frac{B_{2r}\{(1+\beta)^{1-2r} - \beta^{1-2r} - 1\}}{2r(2r-1)} t^{2r-1}. \quad (8.4)$$

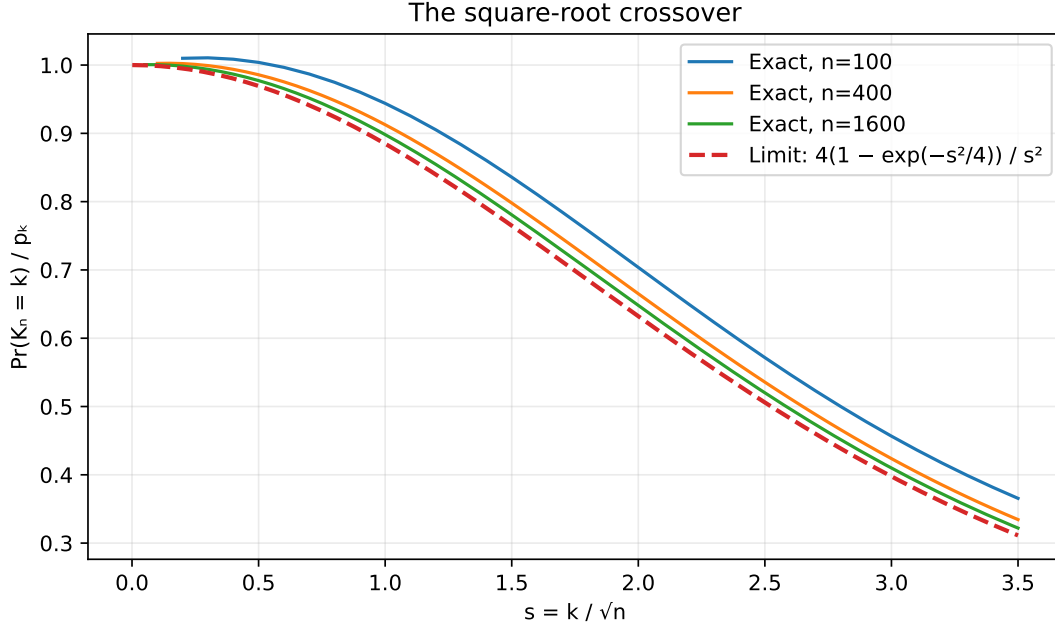


Figure 1: Exact normalized column probabilities approach the crossover function in (7.7). The horizontal coordinate is $k/\sqrt{n}$. These curves show why a fixed- k expansion must not be used uniformly across the whole triangle.

These follow directly from Stirling's formula, using $B_{n,k} = \frac{\beta}{1+\beta} \binom{(1+\beta)n}{\beta n}$. At every finite order the relative remainders are uniform on the stated compact sets.

In particular,

$$R_{n,k} \sim \frac{\sqrt{2}\beta}{\sqrt{1+\beta}} e^{-nJ(\beta)}. \quad (8.5)$$

To see $J > 0$, note that $J(1) = 0$ and $J'(\beta) = \log(2\beta/(1+\beta)) < 0$ for $0 < \beta < 1$. Near the endpoint,

$$J(\beta) = \frac{(1-\beta)^2}{4} + O((1-\beta)^3), \quad (8.6)$$

which explains the square-root crossover from the exponential-sector side.

Remark 8.1 (What an exponentially small term does and does not mean). Equations (8.2)–(8.3), together with the exact identity $T = A - B$, provide a two-sector asymptotic description. A finite inverse-power approximation to A has an error much larger than B . Thus it would be incorrect to append the leading B term to an arbitrarily truncated A series and claim the remainder is smaller than B . To resolve the second sector numerically, retain A exactly through its binomial or gamma expression, or control its approximation to exponentially small accuracy. Here both sectors are defined exactly before either is expanded; no resurgence or optimal-truncation theorem is being assumed.

8.2 Fixed distance from the diagonal

For an integer $\ell \geq 1$ fixed and $n > \ell$, the exact formula specializes to

$$T(n, n - \ell) = 2^{n-\ell-1} \binom{2\ell}{\ell-1} - \binom{n+\ell-1}{\ell-1}. \quad (8.7)$$

The second term is a polynomial in n of degree $\ell - 1$. This is a literal exponential-minus-polynomial expression, not just an asymptotic equivalent. For example,

$$T(n, n - 1) = 2^{n-2} - 1, \quad T(n, n - 2) = 2^{n-1} - (n + 1).$$

The diagonal itself is 2^{n-1} . The formulas are valid with their stated ranges; coincidences with the first zero column at small n are handled without extra exceptions.

9 Critical weighting: exact normalization and a beta law

9.1 An exact critical partition function

For $y > 0$ define a tilted measure by

$$\mathbb{P}_{n,y}(K_n = k) = \frac{T(n, k)y^k}{Z_n(y)}. \quad (9.1)$$

The measure at $y = 1$ is uniform on $\text{Av}_n(123)$. The value $y = 2$ exactly balances the factor 2^{-k} visible in the asymptotics, so it requires a separate analysis.

Theorem 9.1 (Critical normalization and mean). *For $n \geq 1$,*

$$Z_n(2) = (n + 2) \binom{2n}{n} - 4^n, \quad (9.2)$$

$$\mathbb{E}_{n,2} \frac{K_n}{n} = \frac{2}{3} + \frac{4^n}{6Z_n(2)}. \quad (9.3)$$

In particular,

$$Z_n(2) = \frac{4^n \sqrt{n}}{\sqrt{\pi}} \left(1 - \frac{\sqrt{\pi}}{\sqrt{n}} + \frac{15}{8n} - \frac{31}{128n^2} + \frac{21}{1024n^3} + O(n^{-4}) \right), \quad (9.4)$$

$$\mathbb{E}_{n,2}(K_n/n) = \frac{2}{3} + \frac{\sqrt{\pi}}{6\sqrt{n}} + O(n^{-1}). \quad (9.5)$$

The partition function has a full expansion obtained by continuing the central-binomial Stirling series in (9.2).

Proof. In terms of $z = \sqrt{1 - 4x}$, (6.3) at $y = 2$ reduces to

$$F(x, 2) = -1 + \frac{3}{2z} - \frac{1}{z^2} + \frac{1}{2z^3}. \quad (9.6)$$

Use

$$[x^n]z^{-1} = \binom{2n}{n}, \quad [x^n]z^{-2} = 4^n, \quad [x^n]z^{-3} = (2n + 1) \binom{2n}{n}$$

for $n \geq 1$ to obtain (9.2). Differentiating the exact bivariate function gives

$$y \frac{\partial F}{\partial y} \Big|_{y=2} = -\frac{1}{2z} + \frac{1}{2z^2} - \frac{1}{2z^4} + \frac{1}{2z^5}. \quad (9.7)$$

Since $[x^n]z^{-4} = (n+1)4^n$ and $[x^n]z^{-5} = \frac{(2n+1)(2n+3)}{3} \binom{2n}{n}$, coefficient extraction gives

$$\sum_{k=1}^n k 2^k T(n, k) = \frac{2n(n+2)}{3} \binom{2n}{n} - \frac{n}{2} 4^n.$$

Divide by $nZ_n(2)$ and rearrange to prove (9.3). Finally, substitute

$$\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}} \left(1 - \frac{1}{8n} + \frac{1}{128n^2} + \frac{5}{1024n^3} + O(n^{-4}) \right)$$

in the exact expressions. The term -4^n is responsible for the half-power correction in (9.4). $\square$

9.2 An elementary positive-measure comparison

To study the entire distribution, use the same binomial expressions in (7.1) also for $k = 1 < n$, where $A_{n,1} = B_{n,1}$. Extend the positive first term to the diagonal by $A_{n,n} = 2^{n-1}$, and put $B_{n,n} = 0$. Set

$$b_0 = 1, \quad b_\ell = 4^{-\ell} \binom{2\ell}{\ell-1} \quad (\ell \geq 1). \quad (9.8)$$

Writing $\ell = n - k$ gives

$$2^k A_{n,k} = \frac{4^n}{2} b_\ell, \quad 0 \leq \ell \leq n-1. \quad (9.9)$$

These positive weights are close, after normalization, to the actual critical weights $2^k T(n, k)$.

Lemma 9.2. For $N \geq 0$,

$$\sum_{\ell=0}^N b_\ell = -1 + \frac{(2N+1)(N+2)}{N+1} \frac{\binom{2N}{N}}{4^N} = \frac{2\sqrt{N}}{\sqrt{\pi}} - 1 + O((N+1)^{-1/2}), \quad (9.10)$$

where the last estimate is interpreted with a bounded error also at $N = 0$. Moreover,

$$\begin{aligned} \sum_{k=1}^n 2^k A_{n,k} &= (n+1) \binom{2n}{n} - \frac{4^n}{2}, \\ \sum_{k=1}^n 2^k B_{n,k} &= \frac{4^n}{2} - \binom{2n}{n}. \end{aligned} \quad (9.11)$$

Proof. The finite-sum identity in (9.10) is verified at $N = 0$ and then by subtracting its values at N and $N-1$, using the ratio of successive central binomial coefficients. Stirling's formula gives the estimate. Multiply the identity with $N = n-1$ by $4^n/2$ and simplify to get the first equation in (9.11). Subtract (9.2) to get the second. $\square$

Theorem 9.3 (Critical beta limit and error). *Under $\mathbb{P}_{n,2}$,*

$$\frac{K_n}{n} \Rightarrow X, \quad X \sim \text{Beta}(1, \tfrac{1}{2}), \quad f_X(t) = \frac{1}{2\sqrt{1-t}} \quad (0 < t < 1). \quad (9.12)$$

Writing $F_X(t) = 1 - \sqrt{1-t}$ on $[0, 1]$, the distribution functions obey

$$\sup_{0 \leq t \leq 1} |\mathbb{P}_{n,2}(K_n/n \leq t) - F_X(t)| = O(n^{-1/2}). \quad (9.13)$$

All moments converge; in particular the limiting mean is $2/3$ and the limiting variance is $4/45$.

Proof. Let $\tilde{\mathbb{P}}_n$ be the measure on $k = 1, \dots, n$ proportional to $2^k A_{n,k}$. If $A = \sum 2^k A_{n,k}$ and $B = \sum 2^k B_{n,k}$, then $Z_n(2) = A - B$. Since all subtracted weights are nonnegative,

$$\sum_k \left| \frac{2^k T(n, k)}{A - B} - \frac{2^k A_{n,k}}{A} \right| \leq \frac{2B}{A} = O(n^{-1/2}),$$

by (9.11). Thus the two measures differ in total variation by $O(n^{-1/2})$.

Under $\tilde{\mathbb{P}}_n$, the deficit $\ell = n - k$ has weights b_ℓ . The exact partial sum (9.10) and its error imply, uniformly for $0 \leq u \leq 1$,

$$\tilde{\mathbb{P}}_n\{(n - K_n)/n \leq u\} = \sqrt{u} + O(n^{-1/2}).$$

Indeed the numerator is a partial sum through an index within one of un , the denominator is the sum through $n - 1$, and the estimates in (9.10) give the claimed uniform error, including both endpoints. Taking complements and adjusting a possible lattice atom (which is also $O(n^{-1/2})$) gives the beta law and (9.13). Since $K_n/n \in [0, 1]$, weak convergence implies convergence of all moments. The beta density directly gives the stated mean and variance. $\square$

Proposition 9.4 (The first bulk correction). *Uniformly for t in any compact subinterval of $(0, 1)$,*

$$\mathbb{P}_{n,2}(K_n/n \leq t) = F_X(t) + \frac{\sqrt{\pi}}{\sqrt{n}} \left(F_X(t) - \frac{1}{2} \right) + O(n^{-1}). \quad (9.14)$$

This expansion is not asserted uniformly through the endpoints.

Proof. Put $m = \lfloor nt \rfloor$. The positive cumulative weight is $\frac{4^n}{2} \sum_{\ell=n-m}^{n-1} b_\ell$. The constant -1 cancels in the difference of the two partial sums in (9.10). Since both endpoints are proportional to n ,

$$\sum_{k \leq m} 2^k A_{n,k} = \frac{4^n \sqrt{n}}{\sqrt{\pi}} F_X(t) + O(4^n n^{-1/2}).$$

For $k > m$, the ratio bound (7.3) makes the total B weight exponentially small compared with $4^n \sqrt{n}$. Hence

$$\sum_{k \leq m} 2^k B_{n,k} = \frac{4^n}{2} - \binom{2n}{n} + O(4^n e^{-cn})$$

for some $c > 0$ depending on the compact interval (a polynomial factor can be absorbed by reducing c). The numerator is therefore $\frac{4^n \sqrt{n}}{\sqrt{\pi}} F_X(t) - 4^n/2 + O(4^n n^{-1/2})$. Divide by (9.4) to obtain the result. The floor in m changes the leading cumulative expression by only $O(n^{-1})$ after normalization on the stated compact sets. $\square$

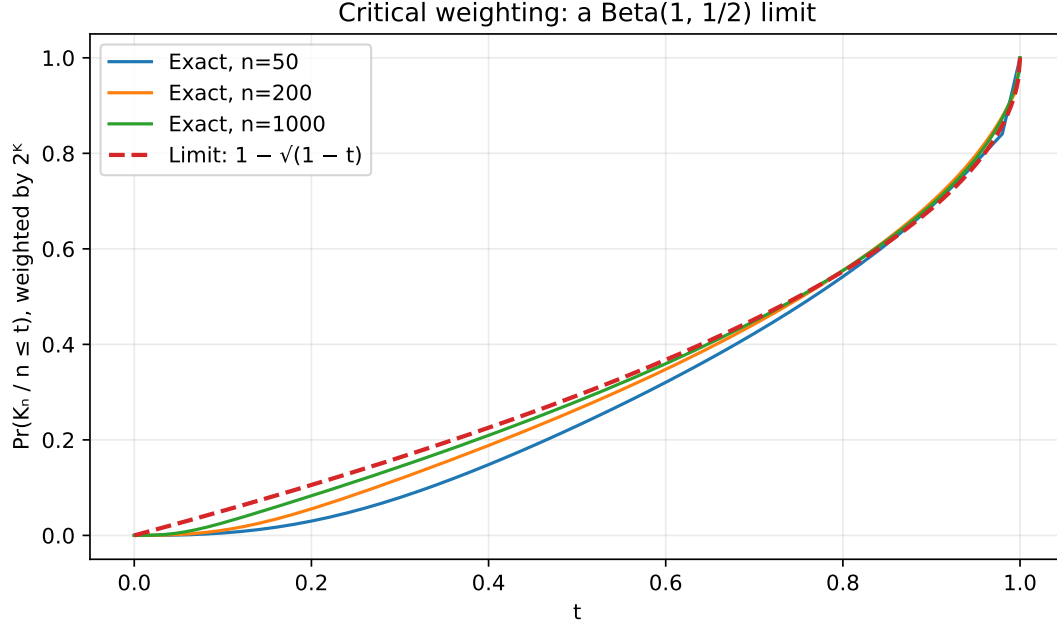


Figure 2: At the critical weight 2^{K_n} , the cumulative distribution of K_n/n approaches $1 - \sqrt{1-t}$. Convergence is only of order $n^{-1/2}$; the exact mean in (9.3) quantifies one part of this finite-size effect.

10 The full tilted phase diagram and its critical window

10.1 Below the transition

Theorem 10.1 (Subcritical law). *For every fixed $0 < y < 2$,*

$$Z_n(y) = P(y)C_n(1 + O_y(n^{-1})), \quad \mathbb{P}_{n,y}(K_n = k) \longrightarrow \frac{p_k y^k}{P(y)}. \quad (10.1)$$

The limiting law has an exponential moment, and $K_n/n \rightarrow 0$ in probability. The partition function has an all-orders inverse-power expansion at each fixed subcritical y .

Proof. The uniform generating-function estimate (6.4) gives the normalization. Combining it with the fixed-column probabilities gives the pointwise limit, whose sum is one by definition of P . Choose a circle $|u| = R < 2$ with $R > y$. The same coefficient estimates give exponentially weighted convergence after multiplying by y^k , so the family is tight and has uniformly bounded moments. Expanding (6.3) to further odd powers of z gives every fixed inverse-power order of the partition function by the same transfer argument. $\square$

10.2 Above the transition

For $y > 2$ put

$$q = \sqrt{1 - 2/y}, \quad \mathcal{A}(y) = \frac{1}{2} \left(1 + \frac{1}{q} - C\left(\frac{1}{2y}\right) \right) = \frac{q^2 + 1}{2q(q+1)}. \quad (10.2)$$

Theorem 10.2 (Supercritical law). *For every fixed $y > 2$, there is $0 < \eta_y < 1$ such that*

$$Z_n(y) = \mathcal{A}(y)(2y)^n(1 + O_y(\eta_y^n)). \quad (10.3)$$

The deficit $D_n = n - K_n$ converges in distribution to a nonnegative integer-valued random variable D_y with

$$\begin{aligned}\mathbb{P}(D_y = 0) &= \frac{1}{2\mathcal{A}(y)}, \\ \mathbb{P}(D_y = \ell) &= \frac{\binom{2\ell}{\ell-1}}{2\mathcal{A}(y)(2y)^\ell}, \quad \ell \geq 1.\end{aligned}\tag{10.4}$$

Its generating function is

$$\mathbb{E}u^{D_y} = \frac{H(u/(2y))}{H(1/(2y))}, \quad H(t) = 1 + (1 - 4t)^{-1/2} - C(t).\tag{10.5}$$

In particular, $K_n/n \rightarrow 1$ in probability.

Proof. The pole $\rho = 1/(2y)$ of $1/(1 - 2xy)$ lies strictly inside the Catalan radius $1/4$. The other denominator in (1.4) is nonzero at ρ , since $\rho y C(\rho) = C(\rho)/2 < 1$. In fact it has no zero in a slightly larger disk: the bound $|xyC(x)| \leq |x|yC(|x|)$ and continuity suffice. The numerator of the simple pole is

$$\lim_{x \rightarrow \rho} (1 - 2yx)F(x, y) = \frac{1}{2} + \frac{C(\rho)^3}{8y\{1 - C(\rho)/2\}} = \mathcal{A}(y).$$

Subtract $\mathcal{A}(y)/(1 - 2yx)$ from F . The remainder is analytic in a disk of radius strictly larger than ρ , so Cauchy's coefficient bound proves (10.3).

For fixed $\ell \geq 1$, multiply (8.7) by $y^{n-\ell}$ and divide by (10.3). The polynomial term tends to zero relative to the exponential term, and the limit is (10.4). The diagonal gives the mass at zero. Finally,

$$\sum_{\ell \geq 1} \binom{2\ell}{\ell-1} t^\ell = (1 - 4t)^{-1/2} - C(t),$$

so the limiting masses sum to $H(1/(2y))/(2\mathcal{A}(y)) = 1$. This proves tightness, the asserted law, and its generating function. $\square$

The limiting deficit has an exponential tail since $1/(2y) < 1/4$. At fixed $y > 2$, moments converge as well: for $u > 1$ sufficiently close to one, $y/u > 2$ and

$$\mathbb{E}_{n,y} u^{D_n} = u^n \frac{Z_n(y/u)}{Z_n(y)} \longrightarrow \frac{\mathcal{A}(y/u)}{\mathcal{A}(y)} = \frac{H(u/(2y))}{H(1/(2y))}$$

uniformly in a neighborhood of $u = 1$.

10.3 Free energy and a finite-size window

Corollary 10.3. *The limiting exponential rate is*

$$f(y) := \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_n(y) = \begin{cases} \log 4, & 0 < y \leq 2, \\ \log(2y), & y \geq 2. \end{cases}\tag{10.6}$$

As a function of $\log y$, its one-sided derivatives at $y = 2$ are 0 and 1. At exactly $y = 2$, the finite-size mean fraction instead converges to $2/3$, in accordance with the nondegenerate beta law.

This derivative jump is the precise sense in which the model has a first-order transition. It is a description of the proved weighted combinatorial model, not a claim about a physical system.

Theorem 10.4 (Critical window). *Let $\lambda \in \mathbb{R}$ be fixed and $y_n = 2e^{\lambda/n}$. Then*

$$\begin{aligned} \frac{Z_n(y_n)}{Z_n(2)} &= M(\lambda) + O_\lambda(n^{-1/2}), \\ M(\lambda) &= \int_0^1 \frac{e^{\lambda t}}{2\sqrt{1-t}} dt = e^\lambda \int_0^1 e^{-\lambda u^2} du. \end{aligned} \quad (10.7)$$

Under $\mathbb{P}_{n,y_n}$, K_n/n converges to the density

$$f_\lambda(t) = \frac{e^{\lambda t}}{2M(\lambda)\sqrt{1-t}}, \quad 0 < t < 1. \quad (10.8)$$

Both the normalization error and the distribution-function error are $O(n^{-1/2})$ uniformly for λ in a compact real interval.

Proof. The normalization ratio is exactly $\mathbb{E}_{n,2} \exp(\lambda K_n/n)$. Apply Theorem 9.3; integration by parts bounds the difference of expectations by the distribution-function error times the total variation of $e^{\lambda t}$ on $[0, 1]$. This variation is uniformly bounded for λ in a compact interval. The substitution $t = 1 - u^2$ gives the second integral. For the tilted distribution functions, apply the same argument to $e^{\lambda t} \mathbf{1}_{t \leq a}$ and divide by the normalization, which remains uniformly bounded away from zero. This proves the stated limit and rate. $\square$

When $\lambda > 0$ one may write $M(\lambda) = e^\lambda \sqrt{\pi} \operatorname{erf}(\sqrt{\lambda})/(2\sqrt{\lambda})$; the integral form is real and nonsingular for every real λ , with $M(0) = 1$.

11 Asymptotic inversion and integer thresholds

11.1 A natural smooth continuation

A discrete sequence does not have a unique real inverse without choosing an interpolation. For fixed $k \geq 2$ we use the gamma continuation of the exact binomial formula:

$$\begin{aligned} T_k(x) &= \frac{2^{k-1} \Gamma(2x - 2k + 1)}{\Gamma(x - k) \Gamma(x - k + 2)} \\ &\quad - \frac{\Gamma(2x - k)}{\Gamma(x - k) \Gamma(x + 1)}, \quad x > k. \end{aligned} \quad (11.1)$$

It equals $T(n, k)$ at all integers $n > k$. The product formula (7.2), now with $\ell = x - k > 0$, proves positivity because each factor is less than one. Differentiated Stirling expansions show that

$$\log T_k(x) \sim ax - b \log x + \log A_k^* + \sum_{j \geq 1} \ell_j(k) x^{-j}, \quad (11.2)$$

as an all-orders asymptotic expansion, where

$$a = \log 4, \quad b = \frac{3}{2}, \quad A_k^* = \frac{pk}{\sqrt{\pi}}, \quad \ell_1 = c_1, \quad \ell_2 = c_2 - \frac{1}{2}c_1^2.$$

Its derivative is $a - b/x + O(x^{-2}) > 0$ for all sufficiently large x . Thus sufficiently large targets have a unique inverse on the eventual increasing branch.

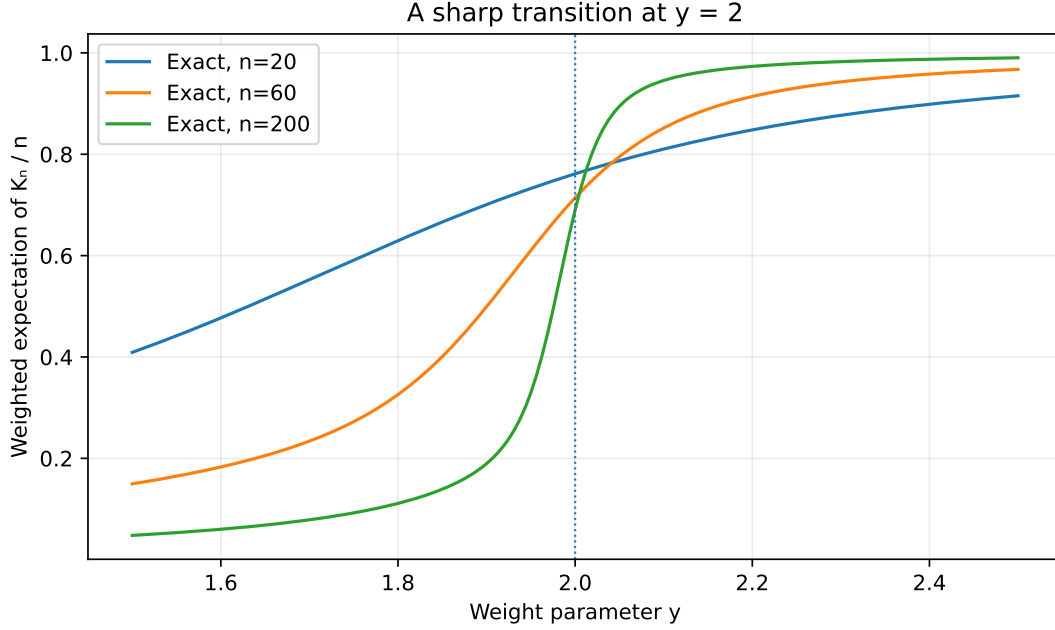


Figure 3: Exact tilted means for increasing n . The limiting mean fraction is 0 for fixed $y < 2$ and 1 for fixed $y > 2$, with the distinct critical value $2/3$. The window $y = 2e^{\lambda/n}$ resolves the transition rather than replacing it by a discontinuous finite- n approximation.

11.2 Lambert- W reduction and corrections

For a target $v \rightarrow \infty$, put

$$Y = \log(v/A_k^*), \quad N = -\frac{b}{a} W_{-1} \left(-\frac{a}{b} e^{-Y/b} \right). \quad (11.3)$$

The real branch W_{-1} , not the principal branch, is required to make $N \rightarrow \infty$. The defining identity for W implies $aN - b \log N = Y$; standard branch conventions are recorded in DLMF §4.13 [6].

Theorem 11.1 (Inverse expansion). *Let $x(v)$ be the eventual increasing inverse of T_k . Then*

$$x(v) = N - \frac{\ell_1}{aN} - \frac{\ell_2/a + b\ell_1/a^2}{N^2} + O_k(N^{-3}). \quad (11.4)$$

An expansion to every order is determined recursively by substituting $\delta = \sum_{j \geq 1} d_j N^{-j}$ in

$$a\delta - b \log(1 + \delta/N) + \sum_{j \geq 1} \ell_j (N + \delta)^{-j} = 0 \quad (11.5)$$

and setting each coefficient equal to zero. At order N^{-j} , the new unknown d_j has coefficient $a \neq 0$.

Proof. Write $x = N + \delta$ in (11.2) and use the exact equation for N . The remaining equation is (11.5). At its first two orders,

$$ad_1 + \ell_1 = 0, \quad ad_2 - bd_1 + \ell_2 = 0,$$

which gives (11.4). Inductively, after the first J coefficients are chosen, the residual in the logarithmic equation is $O(N^{-J-1})$. Its derivative in δ is $a + O(N^{-1})$, bounded away from zero. The mean value theorem

transfers the residual to an inverse error of the same order. Differentiated Stirling remainder bounds justify the derivatives used here [5]. $\square$

[write] Added 1 October 2026, batch 74: an instance of repository results. Theorem 11.1 is the balanced inverse theorem `t2:thm:balanced-inverse` of the combinatorial-inverses volume [7], applied to the gamma continuation (11.1) (a gamma quotient of the kind covered by that volume's engine `t2:thm:gamma`) with $a = \log 4$, $\beta = -b = -\frac{3}{2}$, $C = A_k^*$ and $q_j = \ell_j(k)$. That theorem's $d_1 = -q_1/a$ and $d_2 = \beta q_1/a^2 - q_2/a$ are the two corrections in (11.4); it also gives a nonrecursive formula for every coefficient, and its branch rule (W_{-1} for $\beta < 0$) is the one used here. The integer-threshold identity $Q_k(v) = \lceil x(v) \rceil$ and the bracket (11.7) below are parts (1) and (2) (exact rounding and the separation condition) of the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [8]. The tail inversion (11.9)–(11.10) is a Lambert core of the same kind. No novelty is claimed for this section.

The Lambert quantity itself begins

$$N = \frac{Y + b \log Y - b \log a}{a} + O\left(\frac{\log Y}{Y}\right). \quad (11.6)$$

Repeated substitution, or the standard expansion of W_{-1} , expands it in powers of Y^{-1} with polynomial coefficients in $\log Y$. Keeping N in Lambert form often gives the more accurate practical approximation at a fixed number of displayed terms.

11.3 The rounding issue

Define the eventual integer threshold

$$Q_k(v) = \min\{n \geq k + 1 : T(n, k) \geq v\}.$$

For sufficiently large v , all earlier finitely many values are below the target and the continuation is increasing, so $Q_k(v) = \lceil x(v) \rceil$. However, taking the ceiling of a *truncated* inverse expansion is not uniformly safe: the exact inverse may lie arbitrarily close to an integer. If a rigorous remainder bound is $|x(v) - x_J(v)| \leq \varepsilon_J$, the valid conclusion is the bracket

$$\lceil x_J(v) - \varepsilon_J \rceil \leq Q_k(v) \leq \lceil x_J(v) + \varepsilon_J \rceil. \quad (11.7)$$

An exact binomial evaluation resolves any remaining one- or two-integer ambiguity. This report proves asymptotic remainder orders; it does not claim that the floating-point demonstrations alone provide explicit certified remainder constants or effective monotonicity cutoffs.

11.4 Inverting the limiting tail

The limiting discrete law also has a useful inverse asymptotic. Extend its exact tail formula to large real m and solve

$$\varepsilon = 2^{-m-2}(m^2 + 5m + 2), \quad \varepsilon \downarrow 0.$$

Put $u = m + 5/2$ and $a_2 = \log 2$. Then

$$\varepsilon = \sqrt{2} \left(u^2 - \frac{17}{4} \right) e^{-a_2 u}. \quad (11.8)$$

The leading solution is

$$U = -\frac{2}{a_2} W_{-1} \left(-\frac{a_2}{2} \sqrt{\frac{\varepsilon}{\sqrt{2}}} \right). \quad (11.9)$$

Expanding the logarithm of (11.8) about $u = U$ gives

$$m = U - \frac{5}{2} - \frac{17}{4a_2U^2} + O(U^{-3}). \quad (11.10)$$

Indeed, the neglected factor contributes $\log(1 - 17/(4U^2)) = -17/(4U^2) + O(U^{-4})$ and the derivative of $2 \log u - a_2u$ is $-a_2 + 2/u$. This determines both the sign and the coefficient of the correction. Further terms follow recursively. Integer quantiles require the same rounding care as in (11.7). This tail inversion concerns the limiting law; for finite n , the exact cumulative formula (4.3) is available and should be preferred when feasible.

12 Independent checks, numerical behavior, and reproducibility

12.1 Three logically separate checks

The archive contains `code/verify.py`, using integer arithmetic for enumeration and `sympy/mpmath` for symbolic and numerical checks. The executed default run performed the following tests.

1. **Direct permutation enumeration.** Every permutation of sizes 1 through 9 was examined: 409,113 permutations in total. A three-term increasing-subsequence test decides 123 avoidance. A separate triple scan finds the largest minimum value of a 132 occurrence, hence $K = n - \max\{\text{minimum of a 132 occurrence}\}$, with an empty maximum of zero. The resulting rows agree with the binomial formula.
2. **Insertion-state dynamic programming.** An implementation of the suffix production rule, carrying a flag for whether the first failure has occurred, checks every row through $n = 120$: 7,260 triangle cells. It verifies Catalan row sums, all cumulative counts, the exact critical partition function, its first two raw moments, the polynomial factorization, and the first-order column recurrence. This does not enumerate from the closed formula being checked.
3. **Symbolic and high-precision checks.** Symbolic reduction verifies equality with the posted generating function, the critical specialization, the limit generating function, and the first three fixed-column corrections. Numerical tables use 100 decimal digits and stable logarithmic evaluation where cancellation is relevant.

All checks passed in the supplied run. These computations are independent support for the proofs, not a substitute for them and not a proof-assistant kernel certificate.

12.2 Illustrative numerical results

Table 1 reports relative errors $\text{approximation}/T(n, k) - 1$. The three approximations retain zero, one, or two inverse-power corrections in (5.8).

Table 2 illustrates the slower critical convergence. The finite mean is always larger than $2/3$ by (9.3); the limiting cumulative probability at $1/2$ is $1 - 1/\sqrt{2} \approx 0.2928932188$.

12.3 Stable evaluation and complexity

The two-binomial expression gives a direct exact evaluation with two binomial coefficients and a power of two. With big integers it is safe from cancellation, although bit complexity grows with the output length. For

k	n	Leading term	Through n^{-1}	Through n^{-2}
2	100	$1.3274 \cdot 10^{-3}$	$7.5743 \cdot 10^{-5}$	$1.4262 \cdot 10^{-6}$
3	100	$6.5417 \cdot 10^{-4}$	$1.1810 \cdot 10^{-4}$	$3.6291 \cdot 10^{-6}$
5	100	$7.3479 \cdot 10^{-3}$	$2.1253 \cdot 10^{-4}$	$6.6022 \cdot 10^{-6}$
8	100	$3.7432 \cdot 10^{-2}$	$-1.7498 \cdot 10^{-4}$	$-2.5042 \cdot 10^{-5}$
2	500	$2.5304 \cdot 10^{-4}$	$2.9808 \cdot 10^{-6}$	$1.1250 \cdot 10^{-8}$
3	500	$1.1176 \cdot 10^{-4}$	$4.6048 \cdot 10^{-6}$	$2.8396 \cdot 10^{-8}$
5	500	$1.4269 \cdot 10^{-3}$	$8.2393 \cdot 10^{-6}$	$5.0570 \cdot 10^{-8}$
8	500	$7.2969 \cdot 10^{-3}$	$-6.0141 \cdot 10^{-6}$	$-1.9069 \cdot 10^{-7}$

Table 1: Fixed-column asymptotic errors from the independently evaluated exact formula. All digits shown are rounded from the included CSV file.

n	$\mathbb{E}_{n,2}(K_n/n)$	$\mathbb{P}_{n,2}(K_n/n \leq 1/2)$
20	0.7614630041	0.1888612232
50	0.7197682165	0.2294764254
100	0.7017725312	0.2503403803
200	0.6902952663	0.2640357949
500	0.6809569222	0.2753274754
1000	0.6765433918	0.2807136187
Limit	$2/3$	$1 - 1/\sqrt{2}$

Table 2: Exact critical probabilities, evaluated with arbitrary-precision arithmetic. The substantial finite-size discrepancy is predicted by the $n^{-1/2}$ correction, rather than evidence against the beta limit.

floating-point evaluation when k is small relative to $\sqrt{n}$, the two leading binomials nearly cancel. The stable form is

$$\log T = \log A + \log(-\expm1(\log R)), \quad (12.1)$$

using `log1p` for the factors in (7.2). The verifier uses this form for the gamma continuation and inverse tests.

For a whole table, the suffix-state dynamic program has a straightforward $O(N^3)$ arithmetic-operation implementation through size N , using suffix sums to aggregate state transitions. This is an operation count, not a bit-complexity bound; the integer sizes increase with N . For an individual fixed column, the first-order recurrence is usually more convenient after an initial value is known.

12.4 Rebuilding the archive

The source is `article.tex`; the PDF is compiled with standard LaTeX packages and references embedded in the source. The figure sources and CSV data are included. From the archive directory:

```
python -m pip install -r requirements.txt
python code/verify.py
python code/make_figures.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verifier writes `data/verification_summary.json` and numerical CSV files. The supplied `verification_log.txt` records the executed run. No network access is needed after dependencies have been installed. `OEIS_update.txt` contains suggested entry amendments; they have not been submitted.

[write] Added 1 October 2026, batch 74: the shipped layout. The archive's directory layout is kept, with two moves: its `requirements.txt` is shipped as `data/requirements.txt` and its Makefile as `code/Makefile`; the delivered PDF is not shipped, and `article.pdf` is a build of this text. The commands above therefore still work from the report directory, with `-r data/requirements.txt`; but `verify.py` rewrites `data/verification_summary.json` and the five CSV files, and `make_figures.py` rewrites `figures/`, so run them on a copy of the directory (the README gives the commands). The verifier needs only `mpmath` and `sympy`; `numpy` and `matplotlib` are for the figures.

13 Formalization boundary and further research

13.1 A concrete path into Provelt

The project can be formalized in layers, with a small exact core rather than an immediate attempt to formalize the entire analytic theory. First define the restriction to the largest k entries and prove its monotonicity with respect to pattern containment. Next formalize deletion and insertion of the minimum as inverse operations, and prove the active suffix lemma. Finite counting then gives the safe-prefix distribution and the first-failure convolution. These are the genuinely combinatorial obligations; a formal manipulation of the conjectured generating function alone would not establish that it counts the specified permutations.

The subsequent exact algebra is well delimited: Catalan powers, the binomial telescoper, the entry and tail formulas, and the polynomial-factor recurrence. Each can be checked without trusting floating-point arithmetic. The critical partition function and its moment identities can then be proved by formal-power-series coefficient extraction.

Finally, analytic modules can establish the Stirling expansion, the uniform coefficient-transfer bound, and the probability limits. The repository's generic asymptotic scales and differential-block framework provide

a natural home for the formal expansions and inverse recursion [9]. Their existence does not discharge the analytic remainder obligations. The exact binomial split is especially valuable because it lets a formal statement distinguish the two exponential sectors without assuming an unproved summability or resurgent interpretation.

13.2 Proposed research questions

The following questions go beyond results established in this report. They are proposed research directions, not assertions that every one is a previously published open problem.

1. A family of first-failure statistics. Replace 132 by $1(r+1)r \cdots 2$ while retaining global 123 avoidance. At a 123-active site the right-hand suffix decreases, so a new occurrence is created precisely when at least r entries remain to the right. Thus safe sites become $j = 0, \dots, r-1$, subject to $j \leq d$. Determine the corresponding first-failure generating functions, explicit entry formulas where possible, and the critical exponents and limiting laws. The safe-state process is no longer the same binary reset process; its analysis is the substantive new step.

2. Joint first-failure and suffix statistics. Retain the active suffix length at the first failing insertion and at the final permutation. Derive a multivariate generating function and determine whether the critical beta law admits a useful joint limit with either suffix statistic. The convolution (3.1) retains the failing state j and is a starting point, but the joint limit is not proved here.

3. A fully uniform higher-order theory. The leading square-root crossover and the bulk critical correction are proved above. Develop higher-order expansions that remain uniform as $k/\sqrt{n}$ tends to zero or infinity, and match them to the fixed-deficit region. Similarly resolve the endpoint layers of the critical cumulative law, where (9.14) is not uniform. Explicit remainder constants would make these expansions directly useful for certified computation.

4. Complex zeros and finite-size scaling. Study the zeros of the polynomial $Z_n(y)$ near $y = 2$ and globally. The critical window suggests that the entire function $M(\lambda) = e^\lambda \int_0^1 e^{-\lambda u^2} du$ should govern local complex scaling. Establish the requisite complex-uniform convergence, locate its zeros, and connect them to the accumulation of zeros of Z_n . Real-window convergence alone does not supply this theorem.

5. The polynomial factors H_k . Determine the geometry and asymptotic distribution of the zeros of H_k as $k \rightarrow \infty$, and seek structural coefficient bases or recurrences that explain them. Positivity on $n > k$ and the leading coefficient are proved here; real-rootedness or any broader zero-location claim is not assumed. A useful outcome could be sharper rational bounds on column ratios.

6. Effective inverse certificates. Make the inverse theorems effective by producing explicit monotonicity cutoffs and remainder constants, then combine interval evaluation of W_{-1} and gamma functions with exact binomial checks. The goal is a certified threshold algorithm, with a proof that a short interval contains the answer, rather than an unqualified rounding of an asymptotic formula.

7. Direct bijections to familiar Catalan objects. Translate the first-failure parameter into a natural Dyck-path or ordered-tree statistic, preferably preserving the two low-column interpretations. The insertion tree already supplies an encoding, but a transparent object-level bijection may explain the two-binomial cancellation and the critical beta law without generating functions.

8. Kernel-checked enumeration and reusable analytic interfaces. Implement the exact layer in Lean and use it as a test case for a reusable interface between finite combinatorial enumeration, rational generating functions over the Catalan extension, and asymptotic statements with explicit remainders. The distinction between exact algebraic identities, formal transseries, and analytic estimates should remain visible in the theorem signatures and audit output.

13.3 Conclusion

The posted conjecture for A273821 is accessible through the first insertion that creates a forbidden pattern. That combinatorial event leads to a positive Catalan convolution, a telescoping binomial formula, and a complete fixed-column asymptotic expansion. It also reveals behavior invisible in the original growth-rate comment: a square-root crossover, a precise separation of exponential sectors, and a weighted transition from bounded first-failure time to bounded final deficit, with a beta law exactly at criticality. The exact formulas, rather than numerical extrapolation, support every asserted asymptotic regime.

A Additional exact identities and coefficients

A.1 The OEIS triangle recurrence

With the boundary conventions stated in Section 1, the proved generating function also verifies the recurrence displayed in the OEIS entry:

$$\begin{aligned} T(n, k) = & 3T(n-1, k-1) - 2T(n-2, k-2) \\ & + T(n, k+1) - 2T(n-1, k) + \mathbf{1}_{k=2}C_{n-2}, \quad n \geq 3, 2 \leq k \leq n. \end{aligned} \quad (\text{A.1})$$

Together with $T(1, 1) = 1$, $T(2, 2) = 2$, and the zero boundaries, it computes rows from right to left. It is a consequence of the counting proof, not an assumption used to derive it. One verification is to insert the binomial expression and use Pascal's identity; another is to multiply (1.4) by the recurrence kernel and compare coefficients.

A.2 A third fixed-column correction

The coefficient generator (5.9) gives

$$c_3(k) = -\frac{2k^7 - 38k^6 + 130k^5 + 270k^4 - 3657k^3 + 26093k^2 - 34725k + 13860}{3072(k+4)}. \quad (\text{A.2})$$

This coefficient is independently regenerated in the symbolic check, rather than merely inserted as a numerical fit. For higher orders, the power sums in (5.2) and the exponential recurrence (5.5) are preferable to increasingly large printed polynomials.

A.3 The second critical raw moment

A second application of $y\partial_y$ gives

$$(y\partial_y)^2 F|_{y=2} = \frac{1}{2z} - \frac{3}{2z^2} + \frac{2}{z^3} - \frac{3}{2z^4} + \frac{1}{2z^5} - \frac{1}{z^6} + \frac{1}{z^7}. \quad (\text{A.3})$$

Consequently,

$$\sum_{k=1}^n k^2 2^k T(n, k) = \frac{2(4n^3 + 23n^2 + 63n + 30)}{15} \binom{2n}{n} - \frac{(n+2)(n+4)}{2} 4^n. \quad (\text{A.4})$$

Together with (9.2) and (9.3), this provides an exact finite- n variance and another independent check on the limiting variance $4/45$ after scaling by n^2 .

B Source and claim ledger

Item	Status and use
A273821 definition and conjecture	Primary OEIS entry retrieved October 1, 2026. The entry labels its generating function conjectural and suggests fixed-column growth 4. The present report proves both claims from the permutation definition.
A000245 and A071718	Known companion columns with the initial exceptions specified in (4.8). Their existing leading asymptotics are acknowledged, not claimed as new.
Catalan insertion machinery	Proved directly in Sections 2–3. General Catalan and singularity-analysis methods are established background.
Stirling and transfer estimates	Standard analytic inputs, cited to DLMF and <i>Analytic Combinatorics</i> ; their hypotheses are checked in the uses above. Coefficients and model-specific deductions are derived here.
Exact formulas and limit laws	Conventional proofs supplied in this report; independent computational checks supplied in the archive. No Lean certificate is claimed.
ProveIt connection	The inspected transseries project supplies methodological context and a possible formalization destination, not an assumed proof of any theorem in this report. Repository tree inspected at commit <code>291f6e6bb4477e0cb2131f1e041b12a9abe363fd</code> .
Novelty search	Targeted identifier/statistic searches and repository searches found no independent proof of the posted conjecture. This is not an exhaustive bibliographic certification.
Proposed extensions	Section 13 distinguishes further research from theorems actually proved. The OEIS amendment file is a draft, not a submitted edit.

References

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- [5] NIST Digital Library of Mathematical Functions, §5.11: *Asymptotic Expansions of the Gamma Function*, including the logarithmic Stirling expansion and remainder estimates. Retrieved October 1, 2026. <https://dlmf.nist.gov/5.11>.
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- [7] ProveIt, *Combinatorial Transseries and Their Inverses*, repository volume `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`, theorems labelled `t2:thm:gamma` and `t2:thm:balanced-inverse`. Not formally verified. Added in writing (batch 74).
- [8] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, repository volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`, theorem labelled `p0:thm:staircase`, which that volume records as partly formalized in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. Added in writing (batch 74).
- [9] Vladimir Reshetnikov, *ProveIt*, transseries project README and repository tree inspected October 1, 2026. <https://github.com/VladimirReshetnikov/ProveIt>. Project path: `Analysis/Transseries/README.md`. The README records the independent transseries library, canonical inversion volume, and combinatorial companion; these are contextual rather than logical dependencies of the proofs here.